Thomas Bonar

Mechanical Engineering Senior (B.S. expected May 2027) | Relocating to Dallas–Fort Worth, TX (May 2027) | U.S. Citizen
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Summary

Mechanical engineering senior and Certified SOLIDWORKS Associate who takes hardware from CAD to a machined, tested part. Leads a 5-person CNC workholding capstone, leads a 30-student water project for 1,500 people in Uganda, and supports maintenance at an FDA-regulated (cGMP) manufacturer. Seeking an entry-level mechanical design, manufacturing, or process engineering role.

Education

San Diego State University — *B.S. Mechanical Engineering, GPA: 3.46/4.0* *Expected May 2027*

- **Coursework:** Mechanics of Materials, Mechanical Systems, Manufacturing, Mechatronics, Automatic Control Systems, Thermodynamics, Fluid Mechanics
- **Certifications:** Certified SOLIDWORKS Associate (CSWA); Six Sigma Foundations & Principles; FE Exam (Mechanical) scheduled June 2027

Technical Skills

CAD & Analysis: SolidWorks (CSWA), SolidWorks Simulation (FEA), Onshape, Fusion 360, engineering drawings, tolerancing, MATLAB, Excel, stress, fatigue, and shaft analysis, trade studies
Manufacturing: Haas CNC mill and lathe, G-code, waterjet, MIG welding, 3D printing, fixture and workholding design, DFM/DFA, prototyping
Operations & Quality: SAP PM, preventive maintenance task lists, SOPs and work instructions, BOMs, spare-parts inventory, cGMP documentation, Six Sigma

Engineering Experience

Grifols Diagnostics Solutions Inc. — *Manufacturing Technical Support Intern* *Jun 2026 – Feb 2027*

- At an FDA-regulated (cGMP) plant, digitized 70 preventive maintenance procedures across 32 manufacturing systems into standardized SAP PM task lists, helping technicians complete PMs faster and giving management better workflow data.
- Organized 1,800+ spare parts across 2 facilities, reducing lost parts and helping technicians find parts faster; updated equipment BOMs and worked closely with Quality Control.

SDSU Senior Design: Modular CNC Workholding (Lock Point) — *Team Lead* *Aug 2026 – May 2027*

- Lead a 5-person team designing a modular fixture system for Haas VF-2YT and VF-4SS mills that holds large, thin, or irregular parts; manage a $1,750 budget and weekly sponsor reviews.
- Led a weighted trade study of 3 concepts (selected design scored 4.15/5) and directed holding-force analysis for vacuum (629 lbf), pull-stud (8,000 lbf), and clamp options against a 0.001 in. repeatability target.

Engineers Without Borders (EWB) — *International Project Lead* *Apr 2026 – Present*

- Rebuilt a stalled 5-year-old project from 6 to 30 active members; lead the design of a borehole and hand pump for 1,500 residents of Mbaala, Uganda; wrote the assessment report for EWB-USA and raised $8,000+.

SDSU Rocket Project — *Propulsion Team Member* *Dec 2025 – May 2026*

- Sized the main rotary shaft for a 30,000 rpm electric fuel pump: MATLAB torque, torsional stress, and safety-factor analysis comparing 5 shaft options.

Design Projects

Aircraft Accessory Gearbox (ME 314, team of 8) — *SolidWorks, gear and shaft fatigue analysis* *Fall 2025*

- Designed a 14-gear, 11-shaft gear train driving 4 engine accessories at 152.56 lbf (limit 260 lbf), with minimum safety factors of 3.29 in bending and 5.16 on the accessory shafts.

Torsion Bar Suspension (ME 314, individual) — *Shaft, spline, lug, and fatigue design* *Fall 2025*

- Designed a hollow steel torsion bar, ANSI B92.1 splines, bearing lug, axle, and lever arm; static safety factor of 15.5 against a 1.15 requirement.